Problem 1

SINGLE DIMENTIONAL (EVEN / ODD) PARITY

AIM:
Implementation of data link framing methods for even and odd parity
LOGIC:
0‟s and 1‟s are appended to the data according to the parity of the data.
PSEUDO CODE:
/* for given data */
1. if data bit is 1 increment count else count is zero /* even
parity */
2. if the count mod 2 is zero then append the next bit as 0
else 1 /* odd parity */
3. if the count mod 2 is zero then append the next bit as 1
else 0

INPUT:
Enter the string
1111101

Enter the choice
1. even parity
2. odd parity
1

OUTPUT:
The string is
11111010

INPUT:
Enter the string
111101
Enter the choice
1. even parity
2. odd parity
2

OUTPUT:
The string is
1111010

Problem 2

CHARACTER COUNT

AIM:
Implementation of data link framing methods for counting characters in a given frame.

LOGIC:

The header in the given frame is by default first frame size including the first field data bits are counted and considered as the first frame and the next field contains the next frame size and so on.

PSEUDO CODE:

1. At the sender side the user is asked to enter the number of frames he want to transmit.
2. Depending upon the input, that many number of frames are taken as input from the user and stored in a 2 by 2 matrix.
3. The length of each frame is calculated and stored in a new array.
4. While out putting the frame, the length of each frame is added to the each frame and finally all the frames are appended and sent as a single string.
5. At the receiver side, the first number is treated as the length of the first frame and the string is extracted and displayed.
6. The next number is treated as the length of the next frame and so on.

At Sender:

INPUT:
Enter the number of frames you want to send:2
Enter the frame:1234
Enter the frame:678

OUTPUT
The transmitted frame is:512344678

At receiver:

INPUT:
Enter data
512344678

OUTPUT:
frame sizes are: 5 4
frames are:
frame 1: 1234
frame 2: 678